

Technology Stack

Date	26 June 2026
Team ID	
Project Name	Smart Lender – Loan Approval Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

The Technology Stack defines the tools, frameworks, and technologies used to develop the Smart Lender – Loan Approval Prediction System. It ensures efficient machine learning, web application development, and reliable deployment.

Technology Stack Details :-

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>HTML5, CSS3</i>	<i>Used to create a simple, responsive, and user-friendly interface for entering loan details and displaying prediction results.</i>
	Backend / Server-Side	<i>Python, Flask</i>	<i>Flask handles user requests, processes input data, loads the trained model, and returns prediction results efficiently</i>

3	Database / Data Storage	<i>CSV Dataset & Pickle (.pkl)</i>	<i>The CSV dataset was used for model training, while the Pickle file stores the trained Random Forest model for prediction.</i>
4	Cloud / Hosting / Deployment	<i>Render</i>	<i>Used to deploy the Flask application online, allowing users to access it through a web browser without running it locally.</i>
5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Git tracks project changes, and GitHub stores the source code for version control and collaboration.</i>
6	Third-Party APIs / Other Tools	<i>Scikit-learn, Pandas, NumPy, XGBoost</i>	<i>Used for data preprocessing, machine learning model development, model evaluation, and prediction.</i>